Exponent Upgrade for All p-Admissible Measures

A candidate full resolution of Remark 8.2 in arXiv:2505.20555

Autonomous proof candidate for human review

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Status. Candidate full proof, likely valid pending expert review. The theorem below removes absolute continuity from Proposition 8.1 of Esmayli–Mishra and proves the precise extension proposed in their Remark 8.2.

1 Source question and exact target

Esmayli and Mishra define $H^{1,r}(\Omega;\mu)$ as the completion of smooth functions for the norm

$$\|f\|_{1,r}^r = \int_\Omega |f|^r \, d\mu + \int_\Omega |\nabla f|^r \, d\mu.$$

Their Question 2.21 asks whether, for $1 \le p < s$, membership of u in $H^{1,p}$, together with $u, |\nabla u| \in L^s$, forces $u \in H^{1,s}$ [1]. The question appears on PDF page 14:

would have to have averages of u on $(x_0 - 1, 0)$ and $(x_0, x_0 + 1)$ to be equal, which is not the case.

In view of locality of removability (Proposition 2.23), the proof is complete. □

In the proof of our main result (Theorem 4.3), starting with a u in $H^{1,s}(\mathbb{R}^n \setminus E; \mu)$, where E is porous, at the first stage of the argument we will succeed in proving that $u \in H^{1,p}(\mathbb{R}^n;\mu)$, whereas we need to prove that $u \in H^{1,s}(\mathbb{R}^n;\mu)$, where $s > p$. This arises the following question:

Question 2.21. Let $1 \le p < s$. Suppose $u \in H^{1,p}(\Omega;\mu)$ and we know that u and $|\nabla u|$ are in $L^s(\Omega;\mu)$. Is it true that $u \in H^{1,s}(\Omega;\mu)$?

In Proposition 8.1 we give an affirmative answer to Question 2.21 when $d\mu = w dx$ for a p-admissible weight. We do not know the full generality under which the answer to Question 2.21 is positive.

2.6. **Locality of removability.** Recall the notion of removability of a compact set $E \subset \Omega$ for the Sobolev space $H^{1,p}(\Omega;\mu)$ from Definition 1.1. Results in this section show the local nature of removability, so that, without loss of generality we may choose any convenient Ω as long as it contains the exceptional set E. Analogous statement should be true for $W^{1,p}$ functions.

Proposition 8.1 of [1] proves this for $d\mu = w\,dx$, where w is a p-admissible weight. Remark 8.2 then asks whether the same theorem holds for every p-admissible measure, without assuming absolute continuity. This is the exact target of the present packet (PDF page 32):

8. Appendix

This section is devoted to the proof of a result that was already used at the end of the proof of Theorem 4.3.

Proposition 8.1. *Fix $p \geq 1$ and $d\mu = w dx$ with a p-admissible weight w on $\mathbb{R}^n$. Suppose $u \in H^{1,p}(Q;\mu)$ for a cube $Q \subset \mathbb{R}^n$. If $u \in H^{1,p}(Q;\mu)$ and for some $s > p$ we have $u \in L^s(Q;\mu)$ and $|\nabla u| \in L^s(Q;\mu)$, then $u \in H^{1,s}(Q;\mu)$.*

This answers Question 2.21 in the affirmative for p-admissible weights. While, the analogous claim for $W^{1,p}$ functions would follow immediately from the definitions, for $H^{1,p}$ functions, our proof of Proposition 8.1 relies on some techniques from analysis on metric spaces. The proof is inspired by the proofs in [FHK99] (see also [HKST15, Theorem 10.3.4]). In fact, one might be able to deduce Proposition 8.1 from [FHK99, Theorem 10] which claims that $H^{1,p}$ is equal to the so-called Poincaré-Sobolev space $P^{1,p}$ under a suitable Poincaré inequality.

Remark 8.2. Proposition 8.1 might be true for p-admissible *measures* as well, i.e. without absolute continuity with respect to the Lebesgue measure, as the proof seems to work in that case as well. Since this would require making sure that that all the lemmata from Section 2 that are appealed to in the proof, explicitly or implicitly, remain valid for p-admissible measures, we content ourselves with p-admissible weights.

We will need the fact that a p-Poincaré inequality implies "a (p,p)-Poincaré inequality" [HKST15, Theorem 9.1.15, Remark 9.1.19]

Here μ is p-admissible in the terminology of [1]: it is a Borel doubling measure on $\mathbb{R}^n$, positive and finite on every cube, and there is C_P such that

$$\fint_A |f - f_A| \, d\mu \leq C_P \operatorname{diam}(A) \left(\fint_A |\nabla f|^p \, d\mu \right)^{1/p} \tag{1}$$

for every cube A and every bounded smooth f on A. As recorded in Section 2.5 of [1], doubling and this Poincaré inequality give uniqueness of the Sobolev gradient. We use that established part of their framework.

2 Proof intuition

The useful part of the source proof is its discrete smoothing. On a mesh of size h, replace u by a smooth partition-of-unity interpolation of its μ-averages over the mesh cubes. Differences of neighboring averages are controlled by the original p-Poincaré inequality. Since $\nabla u \in L^s$, Hölder's inequality turns this into a uniform L^s bound for the gradients of the smoothings.

The remaining issue is convergence of the functions in L^s. Applying an s-Poincaré inequality to u here would assume the regularity one is trying to prove. Instead, doubling supplies the ordinary differentiation theorem and the L^s-bounded Hardy–Littlewood maximal operator. The mesh averages converge to u at almost every point and are dominated by $CM_\mu u$, so they converge strongly in L^s. Reflexivity and Mazur's lemma then close the smooth graph. None of these steps refers to a Radon–Nikodym density, Lebesgue convolution, or a distributional derivative.

3 The result

Theorem 1 (measure-valued exponent upgrade)**.** *Let μ be a p-admissible Borel measure on $\mathbb{R}^n$ in the sense above, let $1 \le p < s < \infty$, and let $\Omega \subset \mathbb{R}^n$ be open. Suppose that $u \in H^{1,p}(\Omega;\mu)$ and that its Sobolev pair satisfies*

$$u \in L^s(\Omega;\mu), \qquad |\nabla u| \in L^s(\Omega;\mu).$$

Then $u \in H^{1,s}(\Omega;\mu)$. Its s-Sobolev gradient is the same μ-almost everywhere as its p-Sobolev gradient.

Thus Proposition 8.1 of [1] remains true for arbitrary p-admissible measures, as proposed in Remark 8.2.

4 Preparatory estimates

We first isolate the only consequence of the Poincaré inequality used in the construction.

Lemma 2 (nearby averages)**.** *Let A, B be cubes with $A \subset B \Subset \Omega$, $\ell(B) \le L\ell(A)$, and let $u \in H^{1,p}(\Omega;\mu)$. Then*

$$|u_A - u_B| \le C_L \operatorname{diam}(B) \left(\fint_B |\nabla u|^p \, d\mu \right)^{1/p}. \tag{2}$$

The same estimate, up to twice the constant, controls $|u_{A_1} - u_{A_2}|$ when two comparable cubes A_1, A_2 lie in such a cube B.

Proof. Doubling gives $\mu(B)/\mu(A) \le C_L$. Hence

$$|u_A - u_B| \le \fint_A |u - u_B| \, d\mu \le \frac{\mu(B)}{\mu(A)} \fint_B |u - u_B| \, d\mu,$$

and (2) follows from (1). For an $H^{1,p}$ function, take smooth approximants in the graph norm and pass to the limit. This is legitimate because $\mu(B) < \infty$, so $L^p(B)$ convergence implies $L^1(B)$ convergence. The two-cube statement follows by inserting u_B. $\square$

We write M_μ for the uncentered Hardy–Littlewood maximal operator over cubes. On a doubling metric measure space, M_μ is bounded on $L^r(\mu)$ for every $r > 1$, and locally integrable functions have μ-almost every point as a differentiation point [2].

Lemma 3 (discrete smoothing on nested cubes)**.** *Let $Q \Subset Q' \Subset \Omega$ be cubes. Suppose $u \in H^{1,p}(Q';\mu)$ and $u, |\nabla u| \in L^s(Q';\mu)$. Then $u \in H^{1,s}(Q;\mu)$, and the two Sobolev gradients agree on Q.*

Proof. For small $h > 0$, take the Euclidean grid $\{Q_{h,k}\}$ of side length h and retain the finitely many cells lying within a fixed multiple of h of Q. They and their fixed dilates lie in Q'. A standard smooth partition construction gives nonnegative $\varphi_{h,k} \in C^\infty(Q)$ such that

$$\sum_k \varphi_{h,k} = 1, \qquad \operatorname{supp} \varphi_{h,k} \subset 3Q_{h,k}, \qquad |\nabla \varphi_{h,k}| \le Ch^{-1}, \tag{3}$$

with overlap bounded only in terms of n. Set

$$u_h(x) = \sum_k u_{Q_{h,k}} \varphi_{h,k}(x), \qquad x \in Q.$$

This is smooth. It has finite $H^{1,s}$ norm by the estimates below, so it is an admissible element in the defining smooth class.

Partition Q measurably among the grid cells. If x is assigned to $Q_{h,i}$, every index active at x is a bounded number of cells away. There is a cube $R_{h,i} \subset Q'$, of side at most Ch, containing $Q_{h,i}$ and all active $Q_{h,k}$; the family $\{R_{h,i}\}_i$ has bounded overlap. From $\sum_k \nabla\varphi_{h,k} = 0$, Lemma 2, and then Hölder's inequality,

$$\begin{aligned} |\nabla u_h(x)| &= \left| \sum_k (u_{Q_{h,k}} - u_{Q_{h,i}}) \nabla\varphi_{h,k}(x) \right| \\ &\leq C \left(\fint_{R_{h,i}} |\nabla u|^p \, d\mu \right)^{1/p} \leq C \left(\fint_{R_{h,i}} |\nabla u|^s \, d\mu \right)^{1/s}. \end{aligned} \tag{4}$$

This identity is valid at every point; in particular, no assumption that grid boundaries are μ-null is hidden here. Raising to the s-th power, integrating over the part assigned to $Q_{h,i}$, and summing gives

$$\|\nabla u_h\|_{L^s(Q;\mu)} \leq C \|\nabla u\|_{L^s(Q';\mu)}. \tag{5}$$

Indeed, the assigned set lies in $R_{h,i}$, and the enlarged cubes have bounded overlap.

It remains to prove strong convergence of the function components without assuming that u already has s-Sobolev regularity. If x is a differentiation point of u, every active $Q_{h,k}$ lies in a cube centered at x of side Ch. Doubling compares the measures of these two comparable cubes. Consequently

$$|u_{Q_{h,k}} - u(x)| \leq C \fint_{B_\infty(x,Ch)} |u(y) - u(x)| \, d\mu(y) \longrightarrow 0.$$

Thus $u_h(x) \to u(x)$ for μ-almost every $x \in Q$. The same comparison gives

$$|u_h(x)| \leq C M_\mu(u \mathbf{1}_{Q'})(x). \tag{6}$$

Here $s > p \geq 1$, hence $s > 1$. The doubling maximal theorem makes the right side an L^s function. Dominated convergence now yields

$$u_h \longrightarrow u \quad \text{strongly in } L^s(Q;\mu). \tag{7}$$

For completeness, realize $H^{1,s}(Q;\mu)$ as the closure in $L^s(Q;\mu) \times L^s(Q;\mu)^n$ of the smooth graph $\{(f, \nabla f)\}$. Since $s > 1$, this closed subspace is reflexive. Equations (5) and (7) show that $(u_h, \nabla u_h)$ is bounded, with function component converging strongly to u. Pass to a weakly convergent subsequence in the closed graph and apply Mazur's lemma to tail convex combinations. We obtain smooth functions v_j and some $G \in L^s(Q;\mu)^n$ such that

$$(v_j, \nabla v_j) \longrightarrow (u, G) \quad \text{in } L^s \times L^{sn}.$$

Therefore $u \in H^{1,s}(Q;\mu)$. Since $\mu(Q) < \infty$, the same convergence holds in the p-graph norm. Uniqueness of the $H^{1,p}$ gradient gives $G = \nabla u$ almost everywhere. $\square$

We finally record the elementary sheaf-to-global fact in a form that makes clear that absolute continuity is irrelevant.

Lemma 4 (local graph closure is global)**.** *Let $1 \leq r < \infty$. Suppose a pair (u, G) belongs locally to the smooth $H^{1,r}$ graph on Ω, and $u \in L^r(\Omega;\mu)$, $G \in L^r(\Omega;\mu)^n$. Then $(u, G) \in H^{1,r}(\Omega;\mu)$.*

Proof. Choose a countable locally finite cover $\{U_j\}$ with $U_j \Subset \Omega$, and a smooth partition of unity $\{\psi_j\}$ with $\operatorname{supp}\psi_j \Subset U_j$. Let $\varepsilon_m \downarrow 0$. Local graph membership lets us choose $f_{m,j} \in C^\infty(U_j)$ so that

$$\|f_{m,j} - u\|_{L^r(U_j)} + \|\nabla f_{m,j} - G\|_{L^r(U_j)} \le \frac{\varepsilon_m 2^{-j}}{1 + \|\psi_j\|_\infty + \|\nabla\psi_j\|_\infty}.$$

The locally finite sum $F_m = \sum_j \psi_j f_{m,j}$, with each product extended by zero off U_j, is smooth on Ω. Since $\sum_j \psi_j = 1$ and $\sum_j \nabla\psi_j = 0$, Minkowski's inequality gives

$$\begin{aligned} \|F_m - u\|_{L^r} &\le \sum_j \|\psi_j(f_{m,j} - u)\|_{L^r}, \\ \|\nabla F_m - G\|_{L^r} &\le \sum_j \big(\|\psi_j(\nabla f_{m,j} - G)\|_{L^r} + \|\nabla\psi_j(f_{m,j} - u)\|_{L^r}\big) \le C\varepsilon_m. \end{aligned}$$

Thus $(F_m, \nabla F_m) \to (u, G)$ in the global graph norm. The assumed global integrability also shows that every F_m has finite graph norm. □

5 Proof of Theorem 1

Let $V \Subset \Omega$. Cover $\overline{V}$ by finitely many cubes Q_j for which there are enlarged cubes $Q'_j \Subset \Omega$. Restriction of a smooth approximating sequence shows $u \in H^{1,p}(Q'_j;\mu)$. Lemma 3 gives $u \in H^{1,s}(Q_j;\mu)$, with gradient equal to the original p-gradient. A finite smooth partition of unity therefore shows that the pair $(u, \nabla u)$ belongs locally to the $H^{1,s}$ graph throughout Ω. Its two components are globally in L^s by hypothesis, so Lemma 4 yields $u \in H^{1,s}(\Omega;\mu)$, with the asserted gradient. □

6 Verification, limitations, and novelty

The proof was audited at the four points where singular measures could matter:

1. Lemma 2 is obtained by graph-norm completion and uses only finite cube measure, doubling, and (1).
2. Equation (4) is an algebraic identity for the smooth partition, so positive measure on grid boundaries causes no loss.
3. Equations (6)–(7) are standard doubling-measure facts and use $s > 1$, including when $p = 1$.
4. Lemma 4 explicitly controls the derivative of the partition of unity and does not use a Lebesgue density.

This theorem does not address Open Problem A.18 cited in [1], which asks whether every p-admissible measure on $\mathbb{R}^n$, $n \ge 2$, is absolutely continuous. Instead, it shows that the exponent upgrade remains valid even without knowing the answer to that problem.

A bounded novelty search was performed through 11 August 2026. The run indexes were searched for arXiv:2505.20555 and the exact exponent-upgrade wording. Exact web/arXiv searches used the wording of Remark 8.2 and combinations of "p-admissible measure", "$H^{1,p}$", "$H^{1,s}$", and "discrete convolution". The final journal version, published 10 December 2025, still contains Remark 8.2. No later resolution or equivalent maximal-function proof was found. Novelty confidence is moderate pending a specialist citation search.

The recommended human-review focus is the off-center differentiation argument in Lemma 3 and the applicability of the source's gradient- uniqueness theorem to its exact smooth-test definition of admissibility.

References

[1] B. Esmayli and R. Mishra, *On removable sets for weighted Sobolev functions*, Potential Analysis **64** (2026), article 18; arXiv:2505.20555v2. doi:10.1007/s11118-025-10270-9.

[2] J. Heinonen, *Lectures on Analysis on Metric Spaces*, Universitext, Springer, New York, 2001.